Applying the Backbone Integration Strategy to the Omninet AI Project

Overview

The Omninet AI project consists of three main subsystems:

- **Kiosk (Edge Device)** – touchscreen UI, AI Assistant (on-device), AI Security Agent, sandboxed browser, local cache
- **Omninet Cloud (Core Services)** – API Gateway, microservices (AI Orchestrator, Payment Processor, Device Manager, Analytics & Monitoring), AI Model Registry, databases, message queues
- **AI Support Center (Web Dashboard)** – human agents augmented by AI agents (Monitoring, Billing, Security Analyst)

The backbone integration strategy builds the system incrementally, starting with the communication infrastructure and progressively adding layers of functionality. Each backbone is tested for functionality, error handling, reliability, and performance before proceeding.

Backbone 0: Communication Infrastructure

Components Integrated:

- Basic networking between Kiosk and Cloud (5G/WiFi6/Fiber)
- API Gateway (basic request routing)
- Message Queues (for asynchronous communication)

Testing Performed:

- Establish connection between a simulated kiosk and the cloud
- Send/receive simple heartbeat messages
- Test network error handling (simulated dropouts, reconnection logic)
- Measure latency and throughput under load
- Verify that offline mode detection works (kiosk detects loss of connection)

Quality Risk Addressed:

Is the underlying network architecture and communication backbone stable and performant enough to support 10,000 concurrent kiosks?

Exit Criteria:

- 99.5% of test messages delivered successfully under simulated load
- Automatic reconnection works within 30 seconds after network outage

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Backbone 1: Core Operations and Services

Components Integrated (added to Backbone 0):

- Payment Processor microservice
- Device Manager microservice
- Basic database (for device registration and session tracking)
- Kiosk now includes local cache and basic payment UI

Testing Performed:

- Register a kiosk with the cloud (Device Manager)
- Process a simple cash payment end-to-end (kiosk → Payment Processor → database)
- Test error handling: invalid payment, database timeout, payment service unavailable
- Verify that session data is cached locally if cloud is unreachable (and synced later)
- Measure response time for payment approval (<5 seconds as per SysRD 040-010-040)

Quality Risk Addressed:

Do the core operations (payment, device management) integrate correctly with the communication layer? Can they handle failures gracefully?

Exit Criteria:

- Payment success/failure correctly recorded in database
- Kiosk enters offline mode gracefully when cloud unreachable
- Performance meets SysRD targets

Backbone 2: AI Services and Security

Components Integrated (added to Backbone 1):

- AI Orchestrator microservice
- AI Model Registry
- AI Security Agent (on kiosk)
- AI Monitoring Agent (in cloud for support center)
- Content filtering logic (basic version)

Testing Performed:

- Kiosk requests an AI Assistant response (e.g., language translation) → AI Orchestrator calls appropriate model → response returned
- AI Security Agent detects a blocked website (pornographic content) and blocks access
- Test that AI Monitoring Agent logs security events and alerts the support center dashboard
- Verify that AI model updates can be pushed from cloud to kiosk (or used via API)
- Test error cases: AI model unavailable, inappropriate content false positive/negative

Quality Risk Addressed:

Do the AI services integrate correctly with the existing backbone? Can the system enforce content control and security while maintaining performance?

Exit Criteria:

- AI Assistant responses returned within 2 seconds
- Blocked sites are correctly filtered
- Security events appear in support center dashboard

Backbone 3: Full Kiosk Functionality and User Interaction

Components Integrated (added to Backbone 2):

- Full touchscreen UI (all screens: Welcome, Payment, Browser)
- All payment methods (cash, card, NFC, QR codes, digital wallets)
- Sandboxed browser with full navigation

- Session management (start, extend, terminate, timeout)
- Localization (20+ languages)

Testing Performed:

- End-to-end user journey: Welcome → select language → choose payment → browse web → buy more time → log out
- Test all payment methods and combinations (including promo codes)
- Verify that session timeout works (60-second warning, then termination)
- Test language switching during session
- Ensure browser sandboxing prevents access to local system
- Measure time to Welcome screen after session end (<10 seconds per SysRD)

Quality Risk Addressed:

Does the fully integrated kiosk software provide a smooth, reliable user experience across all features?

Exit Criteria:

- All critical user journeys completed successfully
- No crashes or major UI glitches
- Performance meets all SysRD timing requirements

Backbone 4: AI Support Center Integration

Components Integrated (added to Backbone 3):

- Full AI Support Center web dashboard
- Human agent interfaces
- AI Billing Agent
- AI Security Analyst (advanced threat detection)
- Real-time kiosk monitoring and control

Testing Performed:

- Support center displays real-time status of all active kiosks
- Agent can drill down into a specific kiosk, view anonymized session data
- Agent can terminate a session manually (with appropriate messaging)
- AI Billing Agent automatically issues refunds for service disruptions
- AI Security Analyst flags anomalous behavior (e.g., rapid-fire URL requests)
- Test that terminated user receives correct refund message
- Verify that all actions are logged and auditable

Quality Risk Addressed:

Does the fully integrated system (kiosks + cloud + support center) work together to enable effective monitoring, control, and customer support?

Exit Criteria:

- Support center can manage 1,000 simulated kiosks simultaneously
- AI agents correctly automate billing and security alerts
- All manual overrides work as specified

Backbone N: Fully Integrated System (Ready for System Test)

Components Integrated:

- All components from Backbones 0–4, fully integrated and tested together
- End-to-end test automation drivers simulate real-world usage

Testing Performed:

- Automated test scripts simulate thousands of concurrent user sessions
- Mix of payment types, languages, AI queries, and network conditions
- Measure system stability, performance, and reliability under load
- Verify that no component fails catastrophically under peak load (10,000 concurrent kiosks)

Quality Risk Addressed:

Does the completely integrated system meet all functional and non-functional requirements? Is it ready for formal system testing and eventual deployment?

Exit Criteria:

- All critical test cases pass
- Performance and reliability metrics meet or exceed SysRD targets
- No high-priority defects remain open

Summary of Backbone Progression

Backbone	Focus	Key Risk
0	Communication infrastructure	Is the network backbone stable?
1	Core operations (payment, device mgmt)	Do core services integrate with network?
2	AI services and security	Does AI integrate correctly? Can it filter content?
3	Full kiosk functionality	Does the complete user journey work?
4	Support center integration	Can agents monitor and control the system?
N (Final)	Fully integrated end-to-end	Is the entire system ready for system test?

This backbone integration strategy ensures that each layer of the Omninet AI system is thoroughly tested before adding complexity, reducing the risk of discovering critical integration issues late in the project.